1. Requirements for IoT Platforms

Major Challenges Facing the IoT Industry

- Long rollout period
  - Slow service rollout
  - Fragmented applications, slow development, and long TTM
- Too many standards
  - Multiple devices make integration difficult
  - Different device/sensor manufacturers and protocols
- Difficult network selection
  - Complex network connections
  - Diverse networks: 2G/3G/4G/NB-IoT
  - Security, real-time, and QoS requirements

Requirements for IoT Platforms

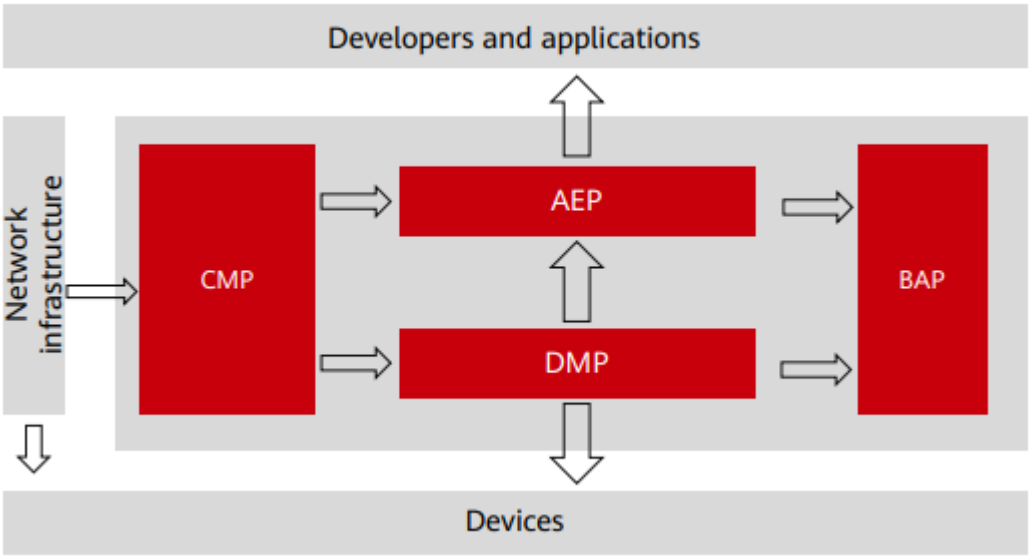
Secure and reliable, supporting device access decoupling and capability openness.

1. Device access decoupling
   - Pre-integrated platform, lightweight development, quick response to market demands
   - Leaving technical details to the IoT platform
2. Secure & reliable
   - Virtualized system, cloud deployment
   - Device access and API authentication
3. Capability openness
   - Open device access
   - Open service application expansion
   - Industry pre-integration and openness management

2. IoT Platform Classification

IoT Platform Classification

Four types: connectivity management platform (CMP), device management platform (DMP), application enablement platform (AEP), business analytics platform (BAP)



3. Open Source IoT Platforms

Open Source:

- Concept: Open source code
- Definition: The source code is open to the public.
- Features: Anyone can view, modify, and use it.

Closed Source:

- Concept: Closed software code
- Definition: The source code is available only to its author.
- Features: Only the author can modify it. Common users only have the right to use the software.

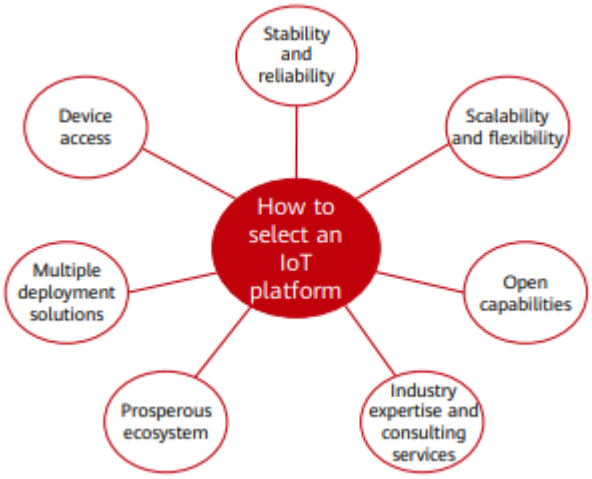
Open Source IoT Platforms

- In wider use than closed source IoT platforms due to: free of charge, low development cost, and open source.
- Popular open source IoT platforms include EMQX, ThingsBoard, and Kaa. Different IoT platforms have differentiated functions to attract users with different requirements.

Comparison of Open Source IoT Platforms

Platform	Product Positioning	Authorization	Pricing	MQTT Broker	Rule Engine	Secondary Development	Language
EMQX	EMQX is an open source IoT MQTT message broker built on the Erlang/OTP platform.	Apache-2.0	Free	√	-	-	Erlang
ThingsBoard	ThingsBoard quickly develops, manages, and extends IoT projects. It provides off-the-shelf IoT cloud or internal solutions and server-side infrastructure for IoT applications.	Apache-2.0	Free	√	√	√	Java
Kaa	Kaa is a highly flexible and scalable platform for building IoT solutions and managing connected devices.	Apache	Free	√	√	√	Java

How to Select an IoT Platform



- Stability and reliability
    - IoT solutions are usually oriented to manufacturing, government, and other commercial scenarios. The stability of an IoT platform directly affects the running of thousands to tens of thousands of devices managed by the IoT platform. An unstable IoT platform consumes a large number of resources for system debugging and O&M, resulting in high costs.
- Scalability and flexibility
    - An IoT project evolves from the proof of concept (PoC) to a large-scale solution. An IoT platform needs to support flexible, fast project deployment and implementation at a low cost in the PoC phase. As the project scale increases, the IoT platform needs to expand functions and improve processing performance to support large-scale services.
- Device access
    - Device connectivity is one of the most fundamental functions of IoT platforms. IoT platforms that support multi-protocol, multi-type IoT devices can meet diverse access requirements. For projects of intelligent reconstruction of existing devices, if an IoT platform has strong access capabilities that support existing devices, the project deployment and R&D costs will be significantly reduced.Huawei Cloud IoT Full-Stack Services

■ IoTDA

IoT Device Access (IoTDA)

IoTDA provides functions such as device fleet access, bidirectional communications between devices and the cloud, batch device management, remote control and monitoring, OTA upgrade, and device linkage rules. It can also flexibly forward device data to other Huawei Cloud services. Using IoTDA, users can quickly connect devices to the platform and integrate applications.

- Method 1: SDK pre-integrated
    - Indirectly connected device: The Huawei AgentLite SDK is integrated on a gateway, and devices that do not have the IP capability are connected to IoTDA through the gateway. This method applies to industrial IoT and smart campus scenarios.
    - Directly connected device: Computing and storage devices that have IP capabilities are directly integrated with the Huawei AgentLite SDK and can connect to IoTDA over HTTPS plus MQTTS.
- Method 2: module integrated
    - Lightweight embedded devices such as sensors, meters, and controllers are integrated with Huawei certified communication modules and can connect to IoTDA using CoAP over LwM2M. This method applies to scenarios that require low power
- Method 3: native protocol access
    - Devices can connect to IoTDA over the native MQTTS protocol. This applies to persistent connection scenarios, such as smart street lamps.

Device Management

IoTDA has extensive device management functions on the console or via calling APIs.

Function	Description
Product model definition	Includes device properties (color, size, collected data, identifiable commands, and reported events). Together, the manufacturer, device type, and device model uniquely identify the device type.
Device registration	Users create a device and configure device information on IoTDA.
Access authentication	IoTDA authenticates connecting devices for data integrity and security, ensuring secure access.
Device access authorization	IoTDA allows an application to grant management rights of its bound devices to another application for easier management.
Device data collection	IoTDA collects device data (service data and alarms) that applications can then subscribe to.
Device shadow	This JSON file stores the reported and expected device status for applications.

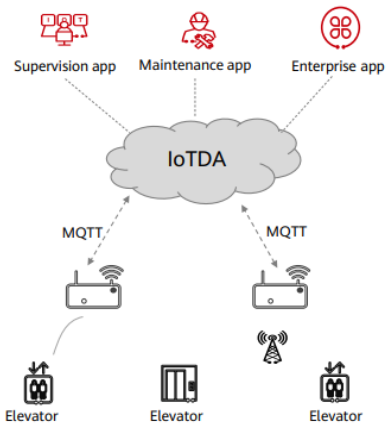
Function	Description
Rule engine	It allows users to set their own rules for devices connected to IoTDA. The devices trigger corresponding actions when conditions are met.
Command delivery	Users deliver commands to remotely control devices through applications or the IoTDA console.
Software and firmware upgrades	Users upgrade software and firmware of devices that support LwM2M and MQTT in over the air (OTA) mode.
Batch operations	Users perform batch operations on devices (device registration, configuration update, command delivery, and software/firmware upgrade).
Device log collection	Users remotely maintain devices by checking device logs collected by IoTDA.
Alarms	Uses manage devices by viewing alarm details and clearing alarms.
Real-time status monitoring	IoTDA monitors device statuses in real time and notifies users of status changes.

Advantages of Huawei Cloud IoTDA

- Simpler device access for all scenarios

- Stable connections and high-concurrency communications
- Full-link self-diagnosis, efficient O&M
- Building cost-effective IoT solutions

IoTDA Application - Smart Elevators



Challenges

• An elevator needs 3–5 sensor types, a significant development workload.

• More connected devices mean more pressure on the platform to maintain performance and scalability.

Benefits

• Multiple sensors are easily connected through an edge gateway that is pre-integrated with a device SDK.

• The platform supports hundreds of millions of connections and millions of concurrent connections, ensuring long-term connectivity for countless devices.

• Elevator data is queried easily and visually for timely O&M and unified supervision.

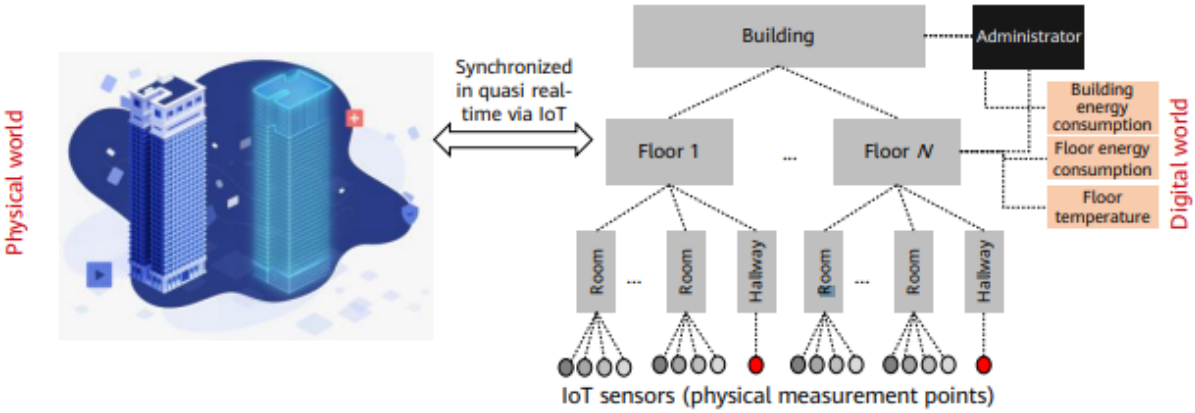
• Elevator control apps and applets for touch-free rides mitigate health risks in public spaces.

■ IoTA

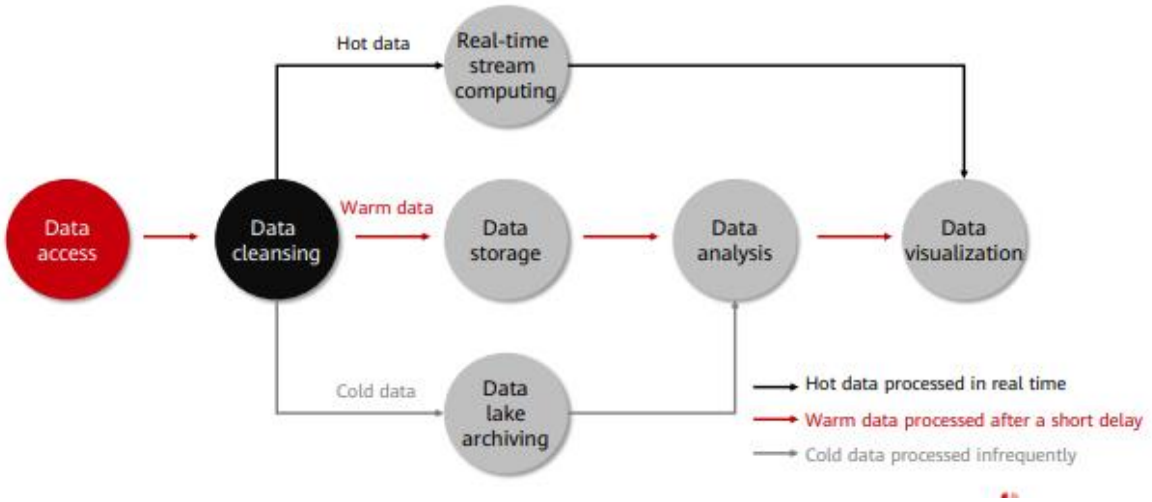
IoT Analytics: Asset Model

Building an asset model is the basis for fully understanding IoT data and building digital twins.

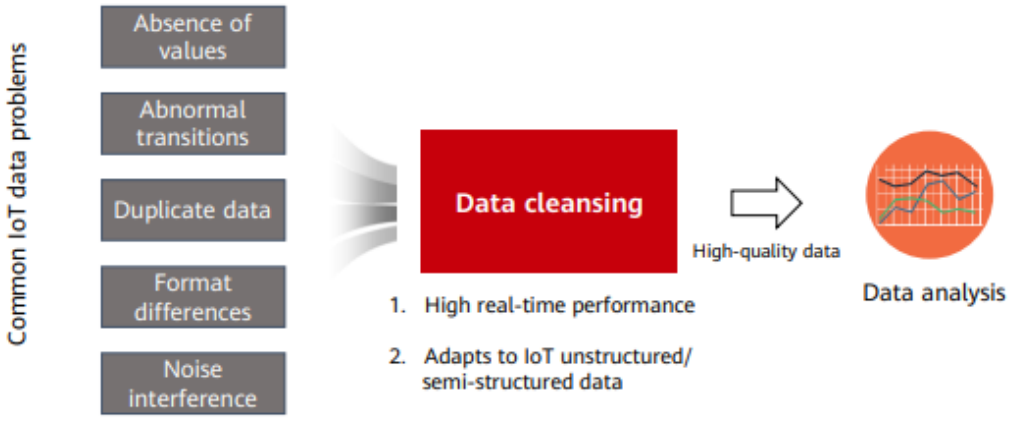
- Establish relationships (between things, between things and space, and between things and people) to understand data in context.
- Use IoT + asset models to build digital twins that are synchronized in quasi-real-time with physical things.
- Model-based abstraction is a unified, service-oriented foundation for data analysis.



IoT Analytics: Multi-temperature Data Management Maximizes Processing Efficiency



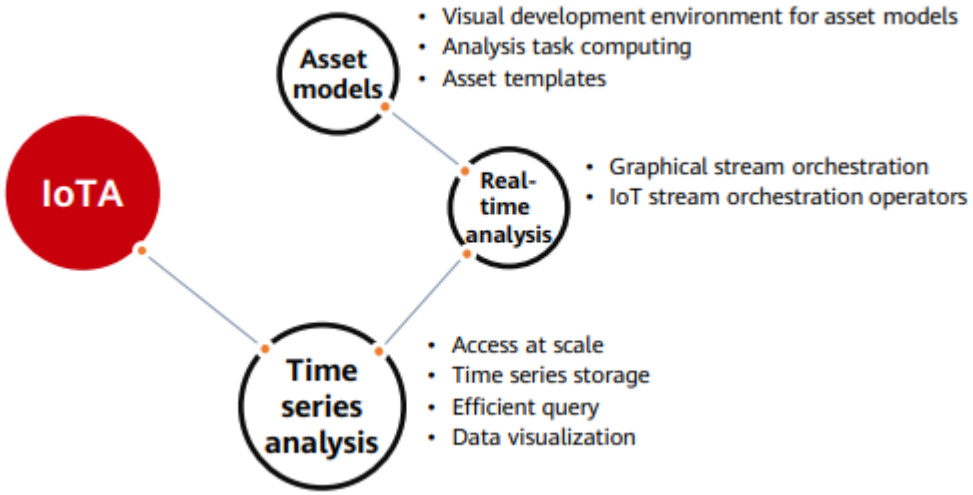
IoT Analytics: Efficient Data Cleansing Provides High-Quality Data for Analysis



IoT Analytics (IoTA)

Streamlines data integration, cleansing, storage, analysis, and visualization using asset models into one-stop services for IoT data developers. It makes development and monetization easier and faster.

IoTA functions:

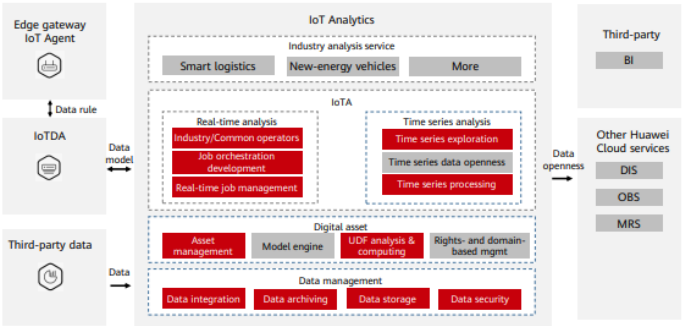


• Asset models: Physical objects or assets can be digitally projected to become computable and interactive. This greatly improves the connection between service systems and the physical world. IoTA helps developers quickly define complex asset models, perform real-time association computing on IoT data, and implement high-performance data access APIs based on these models.

• Real-time analysis: provided based on the big data stream computing engine. To make stream analysis job development easier, IoTA provides graphical stream orchestration so users can quickly develop and roll out services by dragging and dropping components.

• Time series analysis: IoTA provides enhanced time series data processing functions, for example, time series data storage with high compression ratio, efficient query, and a great number of timelines.

IoTA Architecture



IoTA Advantages

- IoT asset models
  IoTA is an analysis service built on asset models. IoTA differs from general-purpose big data cloud services: easy references to IoT model data when defining data analysis jobs to improve job efficiency
- One-stop development
  IoTA uses best practices in big data analysis to provide one-stop data development services. Together with IoTDA, IoTA enables users to develop more efficiently.
- Enhanced time series data processing
  IoTA enhances storage and analysis capabilities for time series data. Example: over 100,000 timelines per instance, data compression ratio up to 20:1, and aggregated computing of data in multiple time dimensions.

IoTA Application Scenarios

- IoT device operations
  Challenges: Enterprises need multi-dimensional statistics and analysis to understand how devices are running. Such data includes quantity growth, activation, time segments when active, and alarms.
  Solution: IoTA seamlessly integrates with IoTDA. Users will receive data sets commonly used in device operations after connecting a device to Huawei Cloud through IoTDA and authorizing IoTA to access the device data. With IoTA, users no longer spend weeks developing data for IoT device operations. Several minutes are enough.
- Enterprise OT data governance
  Challenges: Collected data is poor quality and needs to be cleansed. It costs a lot to store a large amount of OT data or complete data mining. OT data is time-sensitive and degrades quickly.
  Solution: IoTA data pipes and cleansing operators cleanse raw data in drag-and-drop mode. Users also model different types of physical assets and standardize data formats and interactive semantic interfaces. IoTA has a high-performance stream computing engine to process data in milliseconds.
- Smart transportation
  Challenges: Often involves spatiotemporal analysis based on road models. It is challenging to build high-precision models for analysis.
  Solution: IoTA enables users to quickly build computable road models and develop road twins. With these models, users compute the spatiotemporal data from multiple dimensions.

  ■ IoT Stage

This one-stop IoT delivery platform for channel distributors, system integrators, and users facilitates device and application integration, enables industry applications and services, and makes application replication cost-efficient.

- For Sis

IoT Stage is an online experience center with standard profiles and functions such as quick deployment and unified permission management. With IoT Stage, partners upgrade their delivery capabilities, accelerate their deliveries, and expand market potential.

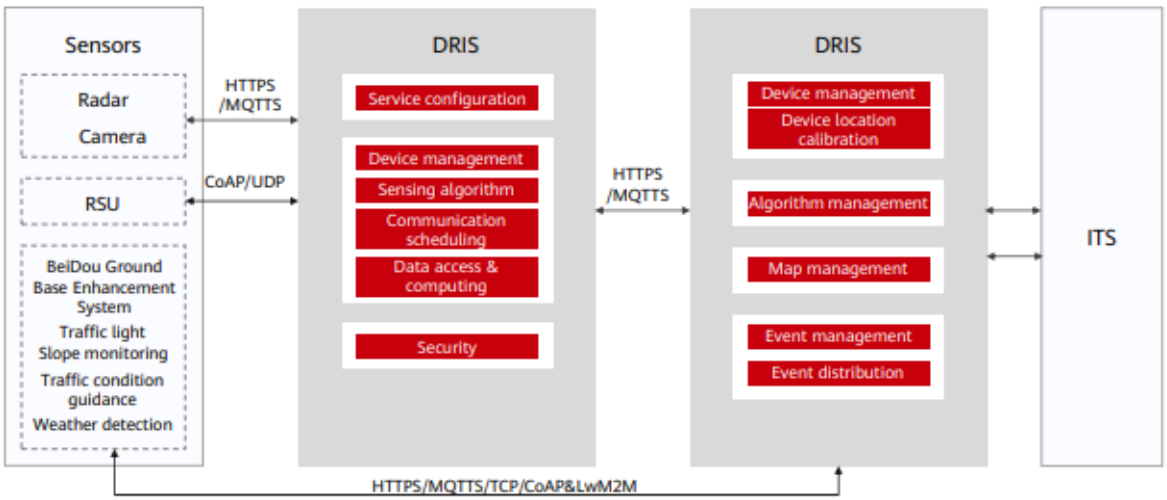
- For application developers
  On the IoT Stage console, developers design the dependency configuration for application deployment, quickly deploy and debug applications, and release them to the IoT marketplace for faster monetization.
- For customers
  IoT Stage is a one-stop purchasing experience. Customers inquire, purchase, and enable applications and cloud resources included in an IoT solution with just a few clicks

■ DRIS

Digital Road Infrastructure Service (DRIS)

- DRIS leverages Huawei IoT services, LTE-V/5G, high-precision positioning, and cloud computing technologies to provide information and facilitate operations.
  Low latency, Millisecond E2E latency with LTE-V and 5G
  All scenarios, 200+ events in standards
  Edge intelligence, High-precision traffic status sensing with Ascend AI processors
  Edge-cloud synergy, Edge intelligent algorithm updates, edge management rule updates, and global status monitoring with open edge computing architecture

DRIS Architecture



• Intelligent Transportation System (ITS) includes IoV technologies such as vehicleto-vehicle and vehicle-road communications. ITS uses dedicated short-range communications (DSRC) and Long Term Evolution-Vehicle (LTE-V) technologies.

• RSU: road side unit

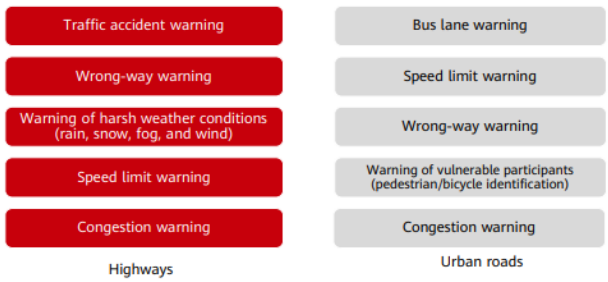
DRIS V2X Server

Function	Description
Data collection	Real-time road events and vehicle/object/traffic light information are aggregated for global data management.
Data openness	Includes third-party platform data and service data (global sensing events, V2X information, and sensor data). Northbound API calling and subscription/push enable other platforms to query and use data.
Device model	Defines roadside device capabilities, with unified device models for devices at the edge.
Device access	Device provisioning, authentication, registration authentication, configuration, data subscription, command delivery, and data storage secure communications between authorized devices and transmitted information.
Precise scheduling	Rules coordinate edge services and long-distance precise push of global events.
Device authentication	Includes one-device-one-secret and two-way certificates, multiple transmission encryption protocols, and EU GDPR compliance.
Event convergence	Analysis of real-time road conditions and events reported by multiple single-point edge computing units generates global events for road sections.
Edge O&M	Edge application versions and configurations are managed together to remotely deploy, upgrade, and monitor applications.
Data storage	Stores system run logs, traffic event information from different channels, and structured data from vehicles.

DRIS V2X Edge

Function	Description
Scenario-specific capabilities	Supports diverse range of vehicle-road collaboration events: warnings of collision, tunnels, incidents, abnormal vehicles, congestion, lane change in blind spots, emergency braking, wrong-way backing and overtaking, parking violations, rapid acceleration/deceleration, sudden turns, speeding, and trucks.
Edge O&M	Remotely deploys event algorithms for standardized management, event capability sensing, and remote O&M and upgrade.
Device model	Defines roadside device capabilities, with unified device models for devices at the edge.
Device access	Ingests data of devices from different vendors that meet industry standards. Such devices include roadside cameras and millimeter-wave radars.
Event analysis	Uses real-time analysis and required event generation to meet E2E industry latency requirements.
Algorithm deployment	Loads in the cloud and deploys on edge computing units, then upgraded and managed.
Communications & distribution	Broadcasts events locally through communication units and reports them to the V2X Server platform.
Sensing convergence	Analyzes data from roadside sensors such as cameras and millimeter-wave radars, and outputs structured data.
Topology management	Manages RSU network topology.

DRIS Applications



- Warning of vulnerable traffic participants (pedestrian/bicycle identification): DRIS monitors real-time locations of pedestrians and bicycles at intersections based on real-time videos and millimeter-wave radars and broadcasts the locations to vehicles nearby. This eliminates blind spots for traffic participants and reduces traffic accidents.
- Wrong-way driving warning: When an abnormal vehicle (for example, a wrongway driving vehicle) appears on the road, DRIS V2X Edge identifies the vehicle and sends a warning signal

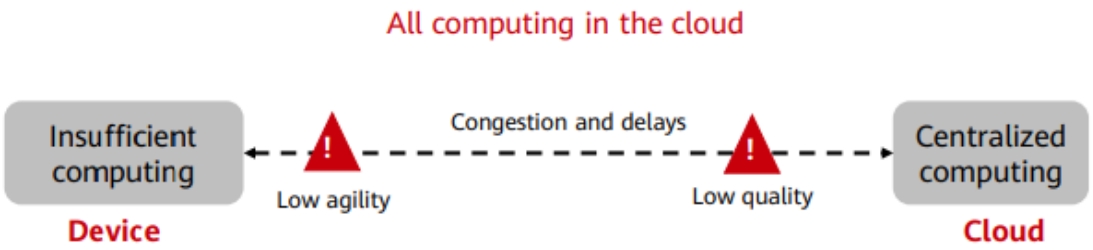
in advance, so oncoming vehicles can decelerate or bypasses the road in advance for accident prevention.

- Speed limit warning: DRIS sends an upcoming speed limit warning signal so oncoming vehicles can decelerate or bypass the road for accident prevention.
- Warning of entering bus lanes: When there is a bus lane on the road, DRIS sends bus lane information in advance. A non-bus vehicle that enters the bus lane will be notified to change a lane to prevent violating rules.
- Congestion warning: When a road is congested, DRIS V2X Edge identifies the road and sends a warning signal in advance, so oncoming vehicles can decelerate or replan the route in advance based on the vehicle-mounted sensor information, thereby improving traffic efficiency.
- Warning of harsh weather conditions such as rain, snow, fog, and wind: DRIS broadcasts harsh weather warnings to traffic participants who will be influenced by the weather in real time based on event delivery or weather analysis, thereby reducing the possibility of traffic accidents.

■ IoT Edge

How is Edge Computing Generated?

The Internet of Everything (IoE) and 5G mean massive growth in network edge devices and data. IoT devices have insufficient computing capabilities, so their data needs to be aggregated for centralized processing. This challenges service timeliness, network stability, and data security. Solution: better edge-cloud collaborative computing and management of edge device applications.



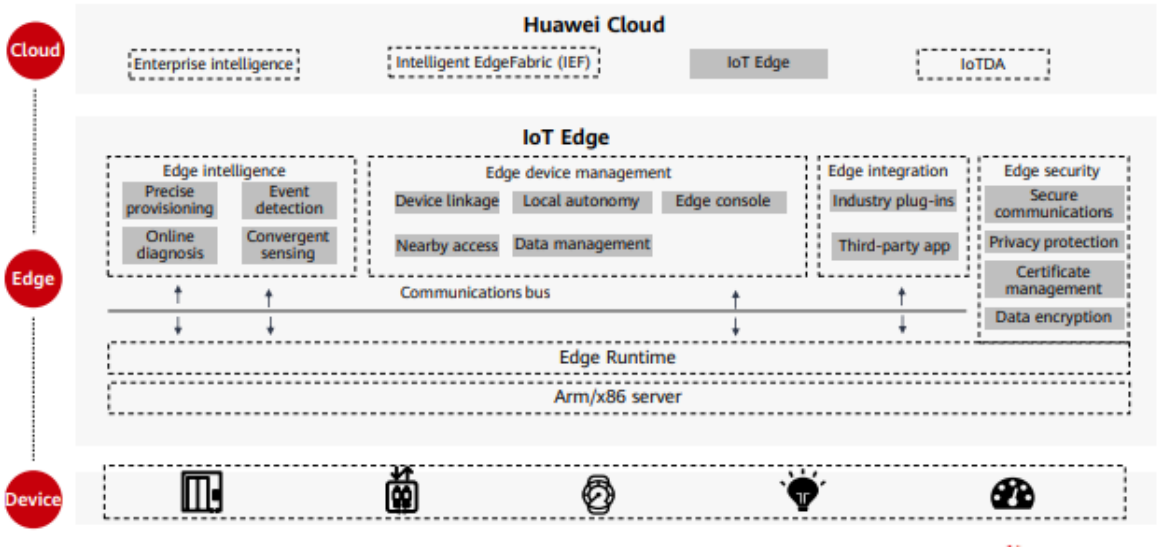
Four Factors for Faster IoT Edge Development

- Low latency: Edge solutions reduce delays as services can be processed closer to where they are needed.
- Massive data: Transmitting heavy data directly to the cloud is expensive, while local data analysis and filtering conserve bandwidth.
- Privacy: Processing enterprise and individual data at the edge ensures enterprise and operations security.
- Local autonomy: Offline processing and self-healing capabilities do not depend on the cloud

IoT Edge is a group of software products, including cloud services, edge runtime software, and edge module applications. It quickly extends cloud capabilities to the edge and provides data collection, low-latency autonomy, cloud-edge collaboration, and edge computing. As the data ingestion point, it meets customers' requirements for device cloud migration, local computing, and data preprocessing in campus, city, and manufacturing scenarios

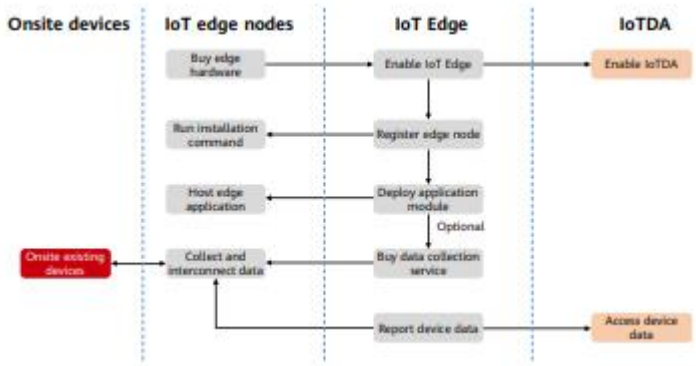
- IoT Edge is an edge computing application for IoT. It is deployed at the edge close to devices or data sources to provide real-time services. Network, compute, storage, and application capabilities make applications intelligent and data secure.
- IoT Edge advantages:
    - Diverse interfaces
    - Hardware
    - Standardized software
    - Application ecosystem

IoT Edge Architecture



- Device side: Devices close to the customer's site can connect to the nearest edge node based on its data collection capability for device management, intelligent control, and data governance.
- Edge node: IoT Edge software is deployed on a gateway or server, which is managed as an edge node. The node implements device data collection, preprocessing, and data routing and forwarding. In addition, the edge side provides application hosting and edge computing, facilitating local autonomy and service expansion.
- Cloud side: The console is used to monitor and manage edge nodes, deliver configurations through edge-cloud synergy channels, deploy and upgrade applications remotely, and route and forward data to the cloud.

IoT Edge Workflow



4. Accessing the Huawei Cloud IoT Platform

Accessing the Huawei Cloud IoT Platform

Access modes:

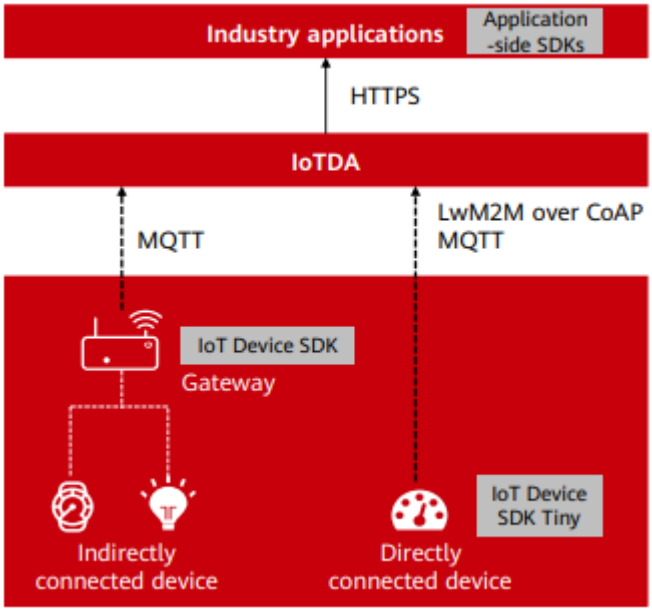
- Using SDKs

  ■ Device access: IoTDA provides IoT Device SDKs for Java (Linux/Windows), C (Linux), C# (Windows), Go Community Edition (Linux/Windows/Unix-like OS), OpenHarmony, and Android.

  ■ Application access: IoTDA provides SDKs for Java, Python, .NET, Go, Node.js, and PHP. π
- Using APIs

  ■ Device access: MQTT, MQTTS, and HTTPS-based APIs

  ■ Application access: HTTP and HTTPS-based APIs



Access using IoT Device SDKs in Different Scenarios

SDK	RAM	Flash	CPU Frequency	Language
IoT Device SDK Tiny	> 32 KB	> 128 KB	> 100 MHz	C

The Device SDK Tiny is for devices with weak computing capabilities.

Water metering, Roadside parking, Smart street light, Gas metering, Bicycle sharing

- MQTT and LwM2M over CoAP
- DTLS/TLS secure transmission protocols
- FOTA and SOTA
- Bootstrap and patterns client, server, and factory supported
- C supported

SDK	RAM	Flash	CPU Frequency	Language
IoT Device SDK	> 4 MB	> 2 MB	> 200 MHz	C, Java, C#, Go, and Android

IoT Device SDKs are for intelligent devices and gateways with strong computing capabilities.

Security monitoring, Child tracking, IoV, Smart home

- MQTT
- Device binding, login, and data reporting
- Child device addition, deletion, modification, and data reporting
- Command receiving
- Java, C, Android, and C# supported
- Node.js, Python, and iOS supported soon

Access using APIs

